

Relational Database Service in AWS

Amazon Relational Database Service (Amazon RDS) is a collection of managed services that makes it simple to set up and scale database in the cloud.

Configuration

Engine type [Info](#)

☐ Amazon Aurora


☒ MySQL


☐ MariaDB


☐ PostgreSQL


☐ Oracle


☐ Microsoft SQL Server


DB instance size

Steps
1
Configure
2
Create
3
Connect
4
Access
5
Monitor
6
Scale
7
Backup
8
Restore
9
Delete

DB instance size

☐ Production db.r6g.xlarge 4 vCPUs 32 GiB RAM 500 GiB 1.017 USD/hour	☐ Dev/Test db.r6g.large 2 vCPUs 16 GiB RAM 100 GiB 0.231 USD/hour	☒ Free tier db.t2.micro 1 vCPUs 1 GiB RAM 20 GiB 0.020 USD/hour
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Relational Database in AWS Types flow Diagram

Task-01

Login to your Amazon Account with username and Password.

Go to the Amazon EC2 console.

Click on "Launch Instance" on the right side of page. Highlight in orange colour.

Choose a "Linux AMI", give a name to your Ec2 instance.

Choose an instance type according to your need, I have select the t2.micro.

Configure the Security group rules to allow inbound traffic on the Appropriate port for the type of database you are using (e.g port 3306 used for MySQL).

The screenshot displays the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information (Mumbai, SaiCloud). Below this is a sidebar with navigation links for Services, S3, EC2, IAM, Console Home, CodeCommit, and AWS Amplify. The main content area shows the 'EC2 > Instances > i-084bc848b201cf4ca' path. The instance summary for 'i-084bc848b201cf4ca (AWS RDS Article)' is displayed, indicating it is 'Running'. Key details include: Instance ID (i-084bc848b201cf4ca), Public IPv4 address (3.110.49.185), Private IPv4 addresses (172.31.3.147), and Public IPv4 DNS (ec2-3-110-49-185.ap-south-1.compute.amazonaws.com). The instance state is shown as 'Running' with a green checkmark icon.

EC2 Dashboard

EC2 Global View

Events

Limits

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Load Balancing

Load Balancers

Target Groups

Instance summary for i-084bc848b201cf4ca (AWS RDS Article) [Info](#)

Updated less than a minute ago

Instance ID

i-084bc848b201cf4ca (AWS RDS Article)

IPv6 address

-

Hostname type

IP name: ip-172-31-3-147.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

3.110.49.185 [Public IP]

IAM Role

-

IMDSv2

Optional

Public IPv4 address

3.110.49.185 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-3-147.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-03db84eb83ec9394d

Subnet ID

subnet-04a0429aa6ec43c16

Private IPv4 addresses

172.31.3.147

Public IPv4 DNS

ec2-3-110-49-185.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

-

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name

-

Details

Security

Networking

Storage

Status checks

Monitoring

Tags

▼ Instance details [Info](#)

Platform

Ubuntu (Inferred)

Platform details

Linux/UNIX

Stop protection

Disabled

AMI ID

ami-03a933af70fa97ad2

AMI name

ubuntu/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-20230328

Launch time

Sat May 20 2023 23:43:07 GMT+0530 (India Standard Time) (3 minutes)

Monitoring

disabled

Termination protection

Disabled

AMI location

amazon/ubuntu/images/hvm-ssd/ubuntu-focal-20.04-amd64-server-

Choose an instance type according to your need, I have select the t2.micro Diagram

aws

Services

Search

[Alt+S]

S3

EC2

IAM

Console Home

CodeCommit

AWS Amplify

Mumbai

SaiCloud

EC2 > Security Groups > sg-0a45af3a9c37bb633 - launch-wizard-19 > Edit inbound rules

Edit inbound rules

[Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

[Info](#)

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0d03cc434eccde96e	HTTP	TCP	80	Custom	
sgr-06c3271b3679dde9d	HTTPS	TCP	443	Custom	

sgr-03d18607905fa2472

SSH

TCP

22

Custom

0.0.0.0/0

Delete

-

MYSQL/Aurora

TCP

3306

Anywhere...

0.0.0.0/0

For Mysql Port Oonnectivity

Delete

Add rule

Cancel

Preview changes

Save rules

In Our Security Group, I will allow the port 3306 used for the MySQL

Launch the instance.

aws

Services

Search [Alt+S]

S3

EC2

IAM

Console Home

CodeCommit

AWS Amplify

Mumbai

SaiCloud

New EC2 Experience

EC2 Dashboard

EC2 Global View

Events

Limits

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Instances (1) Info

Find instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
AWS RDS Article	i-084bc848b201cf4ca	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1b	ec2-3-110-49-185.ap-...

Select an instance

Launch instances

Launch the instance mention a tag on your instance Diagram

2 Create a Free tier RDS instance of MYSQL.

Go to the Amazon RDS console. Click on "Create Database option" highlight in orange colour.

The screenshot shows the Amazon RDS console interface. The top navigation bar includes the AWS logo, a search bar, and links to S3, EC2, IAM, Console Home, CodeCommit, and AWS Amplify. The left sidebar lists various RDS services: Dashboard, Databases, Query Editor, Performance insights, Snapshots, Exports in Amazon S3, Automated backups, Reserved instances, Proxies, Subnet groups, Parameter groups, Option groups, Custom engine versions, Events, Event subscriptions, Recommendations (0), and Certificate update. The main content area features a prominent blue banner with an information icon and the text: "Try the new Amazon RDS Multi-AZ deployment option for MySQL and PostgreSQL. For your Amazon RDS for MySQL and PostgreSQL workloads, improve transactional commit latencies by 2x, experience faster failover typically less than 35 seconds and, get read scalability with two readable standby DB instances by deploying the Multi-AZ DB cluster. Learn more". Below this banner is an orange "Create database" button, which is the element highlighted in the original image. Underneath the banner, it says "Or, Restore Multi-AZ DB Cluster from Snapshot". The "Resources" section displays a table of current RDS resources in the Asia Pacific (Mumbai) region, including DB Instances (0/20), DB Clusters (0/40), Reserved instances (0/20), Snapshots (0), Parameter groups (0), Option groups (0), Subnet groups (0/20), and Supported platforms (VPC). The "Recommended for you" section on the right lists several guides: "Implementing Cross-Region DR", "Test Your DR Strategy in Minutes", "Time-Series Tables in PostgreSQL", and "Migrate SSRS to RDS for SQL Server".

Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Events

Event subscriptions

Recommendations 0

Certificate update

Try the new Amazon RDS Multi-AZ deployment option for MySQL and PostgreSQL

For your Amazon RDS for MySQL and PostgreSQL workloads, improve transactional commit latencies by 2x, experience faster failover typically less than 35 seconds and, get read scalability with two readable standby DB instances by deploying the Multi-AZ DB cluster. [Learn more](#)

Create database

Or, [Restore Multi-AZ DB Cluster from Snapshot](#)

Resources

You are using the following Amazon RDS resources in the Asia Pacific (Mumbai) region (used/quota)

DB Instances (0/20)	Parameter groups (0)
Allocated storage (0 TB/100 TB)	Default (0)
Increase DB instances limit	Custom (0/40)
DB Clusters (0/40)	Option groups (0)
Reserved instances (0/20)	Default (0)
Snapshots (0)	Custom (0/20)
Manual	Subnet groups (0/20)
DB Cluster (0/100)	Supported platforms VPC
DB Instance (0/100)	Default network vpc-03db84eb83ec9394d
Automated	
DB Cluster (0)	
DB Instance (0)	
Recent events (0)	
Event subscriptions (0/20)	

Refresh

Recommended for you

Implementing Cross-Region DR

Learn how to set up Cross-Region disaster recovery (DR) for Aurora PostgreSQL using an Aurora global database spanning multiple Regions. [Learn more](#)

Test Your DR Strategy in Minutes

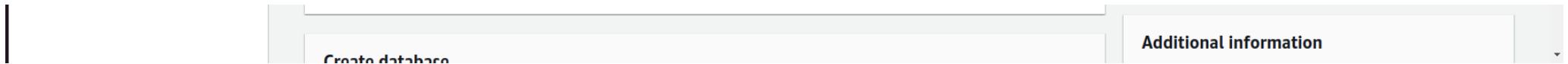
Amazon Aurora Global Database now supports planned managed failover, making disaster recovery drills a breeze. [Learn more](#)

Time-Series Tables in PostgreSQL

Step-by-step guide to design high-performance time series data tables on Amazon RDS for PostgreSQL. [Learn more](#)

Migrate SSRS to RDS for SQL Server

Learn how you can migrate existing SSRS content to an Amazon RDS for SQL Server instance using a PowerShell module. [Learn more](#)



Click on "Create Database option" highlight in orange colour Diagram

In the Database creation method choose the Standard Create.

In Engine option:- I will select the MySQL, you can select according to your need like:- MariaDB, Oracle & PostgreSQL etc.

choose the Standard Create select the MySQL Diagram.

Choose the "Free tier" template for "DB instance class" no charges for the "free tier"

Choose the "Free tier" template for "DB instance Diagram.

Next Step:- Enter a unique name for the "DB instance identifier".

Set the "Master username" & "Master Password" for the database, It will used while login the database from the terminal, I will explain in it in Task.

A screenshot of the AWS Management Console 'Create database' form. The form is titled 'Create database' and is part of the 'Database' section. It shows the 'Standard Create' method selected. The 'Engine' is set to 'MySQL 8.0.32'. The 'DB Instance Identifier' field is filled with 'database-1' and has a red border indicating it is required. Below this field, there is a warning message: 'This field is required'. The 'Manage master credentials in AWS Secrets Manager' checkbox is unchecked. A blue information box states: 'Some features from RDS won't be supported if you want to manage the master credentials in Secrets Manager. Learn more'. The 'Auto generate a password' checkbox is also unchecked. The 'New master password' field is filled with a masked password. The 'Confirm master password' field is also filled with a masked password. The form includes various constraints and links for more information.

Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), ' (single quote), " (double quote) and @ (at sign).

Confirm master password [Info](#)

Enter a unique name for the "DB instance" Set the "Master username" & "Master Password" Diagram.

Set the "Virtual Private Cloud" and "Subnet group" to create the instance in.

Leave the other settings at their default values.

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

Amazon RDS Optimized Writes - new [Info](#)
☐ Show instance classes that support Amazon RDS Optimized Writes

DB instance class [Info](#)
☐ Standard classes (includes m classes)
☐ Memory optimized classes (includes r and x classes)
☒ Burstable classes (includes t classes)

db.t3.micro
2 vCPUs 1 GiB RAM Network: 2,085 Mbps

☐ Include previous generation classes

Storage

Storage type [Info](#)
General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)
20 GiB
The minimum value is 20 GiB and the maximum value is 6,144 GiB

MySQL [X](#)

MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.

- Supports database size up to 64 TiB.
- Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.

"Virtual Private Cloud" and "Subnet group" Diagram.

Storage

Storage type [Info](#)

Allocated storage [Info](#)

MySQL [X](#)

MySQL is the most popular open source database in the world.

Storage

Storage type [Info](#)

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

20

GIB

The minimum value is 20 GiB and the maximum value is 6,144 GiB

Storage autoscaling [Info](#)

Provides dynamic scaling support for your database's storage based on your application's needs.

- ☒ Enable storage autoscaling

Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Maximum storage threshold [Info](#)

Charges will apply when your database autoscales to the specified threshold

1000

GIB

The minimum value is 22 GiB and the maximum value is 6,144 GiB

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

- ☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-03db84eb83ec9394d)

3 Subnets, 3 Availability Zones

Select the Storage Type Diagram.

Select EC2 instance.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, 'Services' link, a search bar, and a keyboard shortcut '[Alt+S]'. Below this is a dark blue bar with icons and labels for S3, EC2, IAM, Console Home, CodeCommit, and AWS Amplify. The main content area is divided into two panes. The left pane shows the 'MySQL' instance configuration page. The 'Connectivity' section is active, displaying a message: 'The minimum value is 22 GiB and the maximum value is 6,144 GiB'. Below this, there's a 'Connectivity' header with an 'Info' link and a refresh icon. Underneath, the 'Compute resource' section explains: 'Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.' The right pane shows the 'MySQL' header and a brief description: 'MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.'

features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.

- Supports database size up to 64 TiB.
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- Supports up to 15 Read Replicas per Instance, within a single Region or 5 read replicas cross-region.

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.



Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.



Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

EC2 instance [Info](#)

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

I-084bc848b201cf4ca

AWS RDS Article



Some VPC settings can't be changed when a compute resource is added

Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 Instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 Instance. You can remove the new security group for the database only by removing the compute resource.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-03db84eb83ec9394d)

3 Subnets, 3 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.



After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Select EC2 instance Diagram.

Choose VPC Security Group:-

Services

[Alt+S]

S3

EC2

IAM

Console Home

CodeCommit

AWS Amplify

Mumbai

SalCloud

DB subnet group name

rds-ec2-db-subnet-group-1

New DB subnet group created.

Public access [Info](#)

☐ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Additional VPC security group

Choose one or more options

MySQL

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- Supports General Purpose, Memory Optimized, and Burstable Performance Instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per Instance, within a single

Additional VPC security group

Choose one or more options ▼

Amazon RDS will add a new VPC security group *rds-ec2-1* to allow connectivity with your compute resource.

Availability Zone [Info](#)

ap-south-1b ▼

Certificate authority - optional [Info](#)

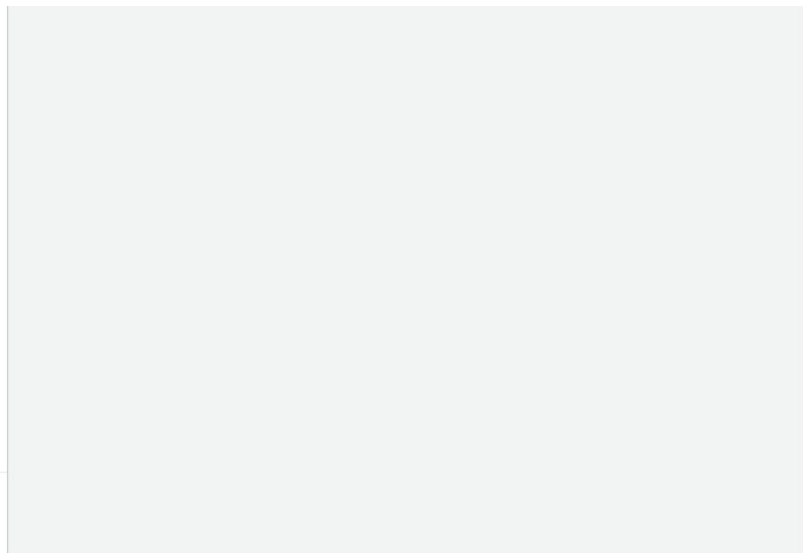
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-2019 (default) ▼

If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port [Info](#)



- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per Instance, within a single Region or 5 read replicas cross-region.

Choose VPC Security Group Diagram.

Click on the "Create database" to start the instance creation.

Services
[Alt+S]

Database authentication

Database authentication options [Info](#)

- ☒ Password authentication
Authenticates using database passwords.
- ☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
- ☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Monitoring

Monitoring

☐ Enable Enhanced monitoring
Enabling Enhanced monitoring metrics are useful when you want to see how different processes or threads use the CPU.

MySQL

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- Supports General Purpose, Memory Optimized, and Burstable Performance Instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per Instance, within a single Region or 5 read replicas cross-region.

► Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Estimated Monthly costs

DB Instance	18.25 USD
Storage	2.62 USD
Total	20.87 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, IOs (if applicable), or data transfer.

Click on the "Create database" Diagram.

aws

Services

Search [Alt+S]

S3 EC2 IAM Console Home CodeCommit AWS Amplify

► Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Estimated Monthly costs

DB Instance	18.25 USD
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Total	20.87 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, IOs (if applicable), or data transfer.

Estimate your monthly costs for the DB Instance using the [AWS Simple Monthly Calculator](#).

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier.](#)

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page](#).

ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

MySQL

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- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per Instance, within a single Region or 5 read replicas cross-region.

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Cancel

Create database

Click on the "Create database" Diagram.

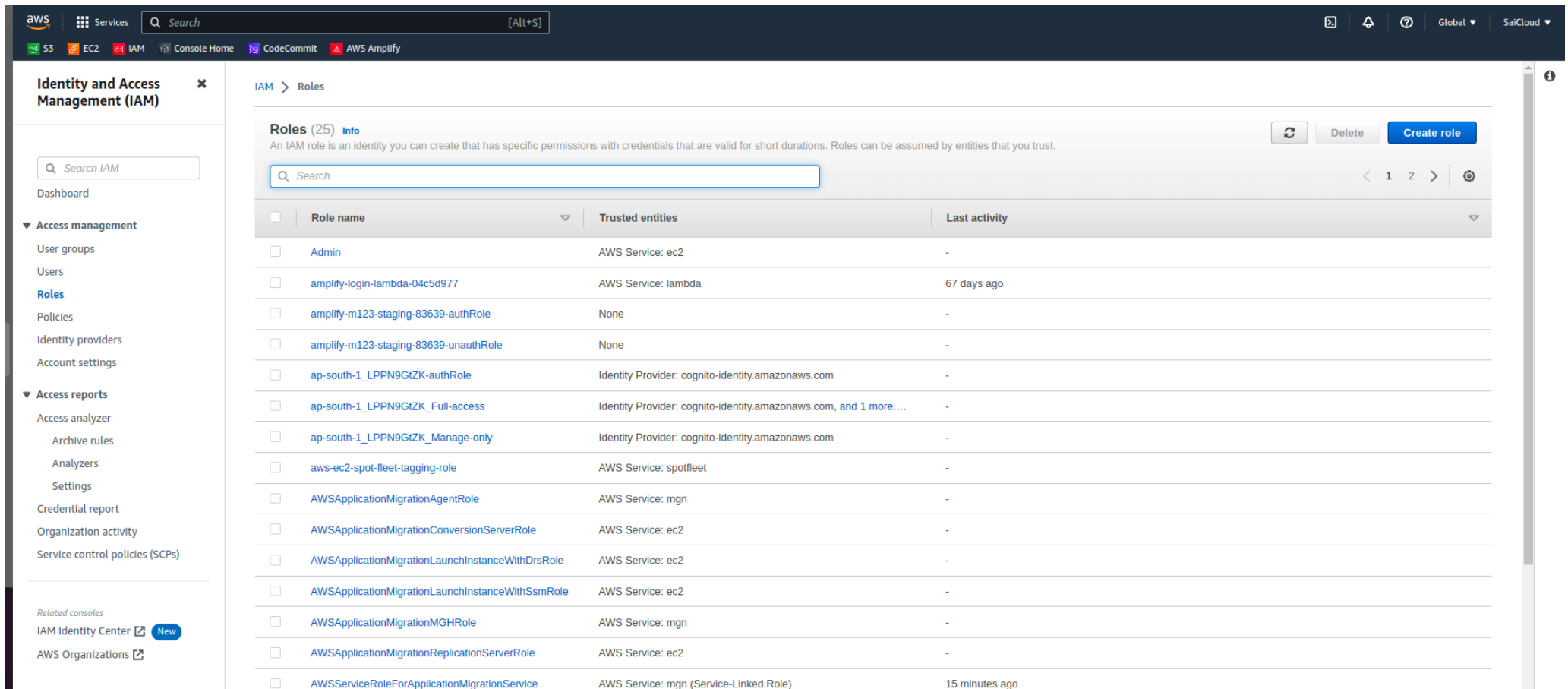
Database is created.

Database is created Diagram.

After completing the Point 1 & Point 2, Move forward to Point to no 3:-

3. Create an IAM role for with RDS access.

Go to the IAM console in AWS. Click on "Roles". click on "Create Role" show in blue colour right side.



The screenshot shows the AWS IAM console interface. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, and Access reports. The main content area displays the 'Roles (25)' page, which includes a search bar, a table of roles, and a 'Create role' button. The table lists various roles with their names, trusted entities, and last activity.

☐	Role name	Trusted entities	Last activity
☐	Admin	AWS Service: ec2	-
☐	amplify-login-lambda-04c5d977	AWS Service: lambda	67 days ago
☐	amplify-m123-staging-83639-authRole	None	-
☐	amplify-m123-staging-83639-unauthRole	None	-
☐	ap-south-1_LPPN9GtZK-authRole	Identity Provider: cognito-identity.amazonaws.com	-
☐	ap-south-1_LPPN9GtZK_Full-access	Identity Provider: cognito-identity.amazonaws.com, and 1 more....	-
☐	ap-south-1_LPPN9GtZK_Manage-only	Identity Provider: cognito-identity.amazonaws.com	-
☐	aws-ec2-spot-fleet-tagging-role	AWS Service: spotfleet	-
☐	AWSApplicationMigrationAgentRole	AWS Service: mgn	-
☐	AWSApplicationMigrationConversionServerRole	AWS Service: ec2	-
☐	AWSApplicationMigrationLaunchInstanceWithDrsRole	AWS Service: ec2	-
☐	AWSApplicationMigrationLaunchInstanceWithSsmRole	AWS Service: ec2	-
☐	AWSApplicationMigrationMGHRole	AWS Service: mgn	-
☐	AWSApplicationMigrationReplicationServerRole	AWS Service: ec2	-
☐	AWSServiceRoleForApplicationMigrationService	AWS Service: mgn (Service-Linked Role)	15 minutes ago

Choose the "AWS Service" Diagram.

Need to be attached the "AmazonRDSFullAccess" policy.

The screenshot shows the AWS IAM console interface. The left sidebar contains the 'Identity and Access Management (IAM)' menu with options like Dashboard, Access management, Access reports, and Related consoles. The main content area is titled 'IAM > Roles > Create role' and shows 'Step 2: Add permissions'. Under 'Permissions policies (Selected 1/848)', a search filter 'rds' is applied, showing one match: 'AmazonRDSFullAccess'. Below this, a table lists the selected policy with columns for Policy name, Type, and Description. The 'Set permissions boundary - optional' section is also visible. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

Policy name	Type	Description
AmazonRDSFullAccess	AWS m...	Provides full access to Amazon RDS via the AWS Management Console.

Need to be attached the "AmazonRDSFullAccess" policy. Diagram

Enter a Unique name for the role.

This block shows the top portion of the AWS IAM console, including the breadcrumb 'IAM > Roles > Create role' and the left sidebar with the 'Identity and Access Management (IAM)' menu.

aws Services Search [Alt+S]

S3 EC2 IAM Console Home CodeCommit AWS Amplify

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings

▼ Access reports

- Access analyzer
 - Archive rules
 - Analyzers
 - Settings
- Credential report
- Organization activity
- Service control policies (SCPs)

Related consoles

- IAM Identity Center [New](#)
- AWS Organizations [New](#)

IAM > Roles > Create role

Step 1
[Select trusted entity](#)

Step 2
[Add permissions](#)

Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+', '@', '-' characters.

Step 1: Select trusted entities [Edit](#)

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    }
15  ]
16 }
```

Step 2: Add permissions [Edit](#)

CloudShell Feedback Language

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Enter a Unique name for the role Diagram.

Click on "Create role"

aws Services Search [Alt+S]

S3 EC2 IAM Console Home CodeCommit AWS Amplify

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings

▼ Access reports

- Access analyzer
 - Archive rules
 - Analyzers
 - Settings
- Credential report
- Organization activity
- Service control policies (SCPs)

Related consoles

- IAM Identity Center [New](#)
- AWS Organizations [New](#)

IAM > Roles > Create role

Step 1
[Select trusted entity](#)

Step 2
[Add permissions](#)

Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+', '@', '-' characters.

Step 1: Select trusted entities [Edit](#)

Step 2: Add permissions [Edit](#)

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Dashboard

▼ Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings

▼ Access reports

- Access analyzer
 - Archive rules
 - Analyzers
 - Settings
- Credential report
- Organization activity
- Service control policies (SCPs)

Related consoles

- [IAM Identity Center](#) **New**
- [AWS Organizations](#)

Step 3

Name, review, and create

Role name

Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+,=,@,-,_' characters.

Description

Add a short explanation for this role.

Maximum 1000 characters. Use alphanumeric and '+,=,@,-,_' characters.

Step 1: Select trusted entities Edit

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    }
15  ]
16 }
```

Step 2: Add permissions Edit

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Click on "Create role" Diagram

IAM Role is Successfully created, I will highlight in Blu Tick.

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Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access management

- User groups
- Users
- Roles**
- Policies
- Identity providers

Role RoleforRD created. View role

IAM > Roles

Roles (Selected 1/26) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

8 matches

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForApplicationMigrationService	AWS Service: mgn (Service-Linked Role)	18 minutes ago
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Linked Role)	27 minutes ago

Roles

Policies

Identity providers

Account settings

▼ Access reports

Access analyzer

Archive rules

Analyzers

Settings

Credential report

Organization activity

Service control policies (SCPs)

Related consoles

IAM Identity Center New

AWS Organizations

☐	AWSServiceRoleForApplicationMigrationService	AWS Service: mgn (Service-Linked Role)	18 minutes ago
☐	AWSServiceRoleForAutoScaling	AWS Service: autoscaling (Service-Linked Role)	27 minutes ago
☐	AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (Service-Linked Role)	3 days ago
☐	AWSServiceRoleForGlobalAccelerator	AWS Service: globalaccelerator (Service-Linked Role)	-
☐	AWSServiceRoleForRDS	AWS Service: rds (Service-Linked Role)	19 minutes ago
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	-
☒	RoleforRD	AWS Service: ec2	-

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Manage

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

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Point No 4 is very Important:-

4 Assign the role to EC2 so that your EC2 instance can connect with RDS.

Go to the Ec2 console.

Select the instance you just created.

Click on "Actions" option then "Instance Settings", then "Attach/Replace IAM Role".

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New EC2 Experience Learn more

EC2 Dashboard

EC2 Global View

Events

Limits

▼ Instances

Instances

Instance Types

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Z
☒	AWS RDS Article	i-084bc848b201cf4ca	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1b

Connect

Instance state

Actions

Launch instances

Connect

View details

Manage instance state

Instance settings

Networking

Security

Image and templates

Monitor and troubleshoot

Change security groups

Get Windows password

Modify IAM role

Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

► Network & Security

▼ Load Balancing
Load Balancers

▼ Instance: i-084bc848b201cf4ca (AWS RDS Article)

Details | Security | Networking | Storage | Status checks | Monitoring | Tags

▼ Instance summary Info

Instance ID i-084bc848b201cf4ca (AWS RDS Article)	Public IPv4 address 3.110.49.185 open address	Private IPv4 addresses 172.31.3.147
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-3-110-49-185.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-3-147.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-3-147.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.110.49.185 [Public IP]	VPC ID vpc-03db84eb83ec9394d	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-04a0429aa6ec43c16	
IMDSv2		

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Click on "Actions" option then "Instance Settings", then "Attach/Replace IAM Role" Diagram

Choose the IAM role you just created.

Click on the "Updated IAM role"

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S3 EC2 IAM Console Home CodeCommit AWS Amplify

EC2 > Instances > i-084bc848b201cf4ca > Modify IAM role

Modify IAM role Info

Attach an IAM role to your instance.

Instance ID
i-084bc848b201cf4ca (AWS RDS Article)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

RoleforRD [Create new IAM role](#)

Cancel Update IAM role

Cancel

Update IAM role

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Choose the IAM role you just created Click on the "Updated IAM role" Diagram.

5 Once the RDS instance is up and running,get the Credentials and connect your EC2 instance using a MySQL client.

Go to RDS console in AWS.

Select the Instance you just created.

Click "Configuration" and note the endpoint address.

The screenshot displays the AWS Management Console interface for the Amazon RDS console. The top navigation bar includes the AWS logo, a search bar, and links to various services like S3, EC2, IAM, and CloudShell. The left sidebar shows the 'Amazon RDS' section with a list of navigation options: Dashboard, Databases (highlighted), Query Editor, Performance Insights, Snapshots, Exports in Amazon S3, Automated backups, Reserved Instances, and Proxies. The main content area shows the 'database-1' instance details. The breadcrumb navigation is 'RDS > Databases > database-1'. The instance name 'database-1' is prominently displayed at the top of the main content area, with 'Modify' and 'Actions' buttons to its right. Below this, a 'Summary' section provides key details about the instance in a table format. At the bottom, a horizontal tab bar allows switching between different views: 'Connectivity & security' (selected), 'Monitoring', 'Logs & events', 'Configuration', 'Maintenance & backups', and 'Tags'.

Summary			
DB Identifier database-1	CPU 2.41%	Status Available	Class db.t3.micro
Role Instance	Current activity 0 Connections	Engine MySQL Community	Region & AZ ap-south-1b

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Events

Event subscriptions

Recommendations 2

Certificate update

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Connectivity & security

Endpoint & port

Endpoint

database-1.c4mwnlabtvwh.ap-south-1.rds.amazonaws.com

Port

3306

Networking

Availability Zone

ap-south-1b

VPC

vpc-03db84eb83ec9394d

Subnet group

rds-ec2-db-subnet-group-1

Subnets

subnet-07aa9f9d6f0be6d5d

subnet-096308b99a7f872ad

subnet-0914c635013f85493

Network type

Security

VPC security groups

rds-ec2-1 (sg-0175f1aeb3760cdda)

Active

Publicly accessible

No

Certificate authority

Info

rds-ca-2019

Certificate authority date

August 22, 2024, 22:38 (UTC+05:30)

DB Instance certificate expiration date

August 22, 2024, 22:38 (UTC+05:30)

Click "Configuration" and note the endpoint address Diagram.

Click "Security" and note the username and Password.

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Parameter groups

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Custom engine versions

Events

Event subscriptions

Recommendations 2

Instance

Configuration

DB Instance ID

database-1

Engine version

8.0.32

DB name

-

License model

General Public License

Option groups

default:mysql-8-0 In sync

Amazon Resource Name (ARN)

arn:aws:rds:ap-south-1:414310061589:db:database-1

Resource ID

db-D4WG6TS6PW5SV426RT2FCSKCLA

Created time

May 21, 2023, 00:02 (UTC+05:30)

DB Instance parameter group

Instance class

Instance class

db.t3.micro

vCPU

2

RAM

1 GB

Availability

Master username

admin

Master password

IAM DB authentication

Not enabled

Multi-AZ

No

Secondary Zone

-

Storage

Encryption

Enabled

AWS KMS key

aws/rds

Storage type

General Purpose SSD (gp2)

Storage

20 GiB

Provisioned IOPS

-

Storage throughput

-

Storage autoscaling

Enabled

Maximum storage threshold

1000 GiB

Performance Insights

Performance Insights enabled

Turned off

Recommendations 2

Certificate update

Created time May 21, 2023, 00:02 (UTC+05:30)	NO Secondary Zone	1000 GiB
DB Instance parameter group default.mysql8.0 In sync		
Deletion protection Disabled		

Recommendations (2)

Dismiss Schedule Apply now

Click "Security" and note the username and Password Diagram.

SSH into your EC2 instance using a Terminal or remote access tool.

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EC2 > Instances > i-084bc848b201cf4ca > Connect to instance

Connect to instance [Info](#)

Connect to your instance i-084bc848b201cf4ca (AWS RDS Article) using any of these options

EC2 Instance Connect Session Manager **SSH client** EC2 serial console

Instance ID
i-084bc848b201cf4ca (AWS RDS Article)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this Instance is apache1.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
chmod 400 apache1.pem
4. Connect to your Instance using its Public DNS:
ec2-3-110-49-185.ap-south-1.compute.amazonaws.com

Command copied

```
ssh -i "apache1.pem" ubuntu@ec2-3-110-49-185.ap-south-1.compute.amazonaws.com
```

Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel

SSH into your EC2 instance Diagram.

Successfully login in account with SSH Key Diagram.

Need to be install the MySQL Client, such as "mysql".I will share the commands it will helpful for you for the installtion.

Syntax:-

```
sudo apt install mysql-client-core-8.0
```

```
mysql --version
```

install the MySQL Client, such as "mysql" & Version check Diagram.

Connect to the RDS instance using the MySql client and the endpoint address username, and Password.

Enter the password when prompter and press enter:-

you should now be connected to the MySQL database on the RDS instance.

Syntax:-

```
sudo mysql -h database-1.c4mwnlabtvwh.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
```

Enter the Password:- <>

you are enter in the mysql:-

```
mysql>
```

```
show databases;
```

```
Last login: Fri May 19 18:18:19 2023 from 106.204.197.71
ubuntu@s3bucket:~$
ubuntu@s3bucket:~$ sudo apt install mysql
mysql-client          mysql-testsuite
mysql-client-8.0      mysql-testsuite-8.0
mysql-client-core-8.0 mysqlcl
```

```

ubuntu@s3bucket:~$ sudo apt install mysql
mysql-client          mysql-testsuite
mysql-client-8.0      mysql-testsuite-8.0
mysql-client-core-8.0 mysqltcl
mysql-common          mysqltuner
mysql-router          mysqlmail
mysql-sandbox         mysqlmail-courier-logger
mysql-server          mysqlmail-dovecot-logger
mysql-server-8.0      mysqlmail-postfix-logger
mysql-server-core-8.0 mysqlmail-pure-ftpd-logger
mysql-source-8.0
ubuntu@s3bucket:~$ sudo apt install mysql
mysql-client          mysql-testsuite
mysql-client-8.0      mysql-testsuite-8.0
mysql-client-core-8.0 mysqltcl
mysql-common          mysqltuner
mysql-router          mysqlmail
mysql-sandbox         mysqlmail-courier-logger
mysql-server          mysqlmail-dovecot-logger
mysql-server-8.0      mysqlmail-postfix-logger
mysql-server-core-8.0 mysqlmail-pure-ftpd-logger
mysql-source-8.0
ubuntu@s3bucket:~$ sudo apt install mysql-client-core-8.0
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  mysql-client-core-8.0
0 upgraded, 1 newly installed, 0 to remove and 17 not upgraded.
Need to get 5173 kB of archives.
After this operation, 75.5 MB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu focal-updates/main amd64 mysql-client-core-8.0 amd64 8.0.33-0ubuntu0.20.04.2 [5173 kB]
Fetched 5173 kB in 1s (6318 kB/s)
Selecting previously unselected package mysql-client-core-8.0.
(Reading database ... 91438 files and directories currently installed.)
Preparing to unpack .../mysql-client-core-8.0_8.0.33-0ubuntu0.20.04.2_a
md64.deb ...
Unpacking mysql-client-core-8.0 (8.0.33-0ubuntu0.20.04.2) ...
Setting up mysql-client-core-8.0 (8.0.33-0ubuntu0.20.04.2) ...
Processing triggers for man-db (2.9.1-1) ...
ubuntu@s3bucket:~$ mysql --version
mysql Ver 8.0.33-0ubuntu0.20.04.2 for Linux on x86_64 ((Ubuntu))
ubuntu@s3bucket:~$

```

`sudo mysql -h database-1.c4mwnlabtvwh.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p Diagram`

(Note) if your RDS are not access please check the Endpoint address in RDS. It will take one or 2 min for access.

In this blog, I have discussed RDS Definition, create a database, attach with the IAM Role . If you have any questions or would like to share your experiences, feel free to leave a comment below. Don't forget to read my blogs and connect with me on LinkedIn and let's have a conversation.

Thank you for reading !! I hope you find this article helpful!!

Happy Learning!!